What we're doing

Collect rollouts from Eric's tuned contacts-v1 1.5B model (eval loss 2.7566) at scale, and learn how fast we can generate them on TPU. Sample 1000 training targets (round-0, $L \leq 512$, ≥ 5 contacts); for each, generate 1000 rollouts; score every rollout's precision/recall/F1; save the best-recall and best-F1 rollouts verbatim for a future "train on high-accuracy rollouts" experiment.

Why

Groundwork for the real question: does fine-tuning on high-accuracy rollouts beat re-epoching the data? First we need the rollouts and a feel for the generation cost at scale. Builds on exp82 (rollout+resample is the best LM-only recipe) and exp89 (the vLLM/iris-TPU path).

How (inference)

vLLM on iris TPU v5p-8, bf16 model, **tensor_parallel_size=4** (all 4 chips). rollout+resample recipe (each rollout a fresh document realization). Prefixes pre-built locally; the TPU worker is marinfold-free and resumes on restart.

Calibration — throughput

On one v5p-8 (10 length-stratified targets × 1000 rollouts): tp=4 gives **~11.3k tok/s aggregate (~16–18k steady-state)**, ~1.8–2× over tp=1 (which left 3 of 4 chips idle). 0% of rollouts hit the token cap. resample ≈ nsample on accuracy, ~15% slower. Projected full 1M-rollout run: ~0.86 h on 8× v5p-8.

Calibration — accuracy

Per-rollout accuracy is low on average but **best-of-1000 ≫ mean**, and strongly length-dependent: small proteins (L≈30–80) reach best recall 0.8–1.0, large ones (L>250) stay near 0.15–0.2. This is the point of sampling many rollouts — the best one is far better than the typical one.

Full run — scale

1,000,000 rollouts (1000 targets × 1000) in **~41 min wall on 8× v5p-8** (~4.6 v5p-8-hours of generation; 283 M tokens; 17.2k tok/s effective; 0% truncated; 0 failures). 10× more targets ≈ ~7.5 h on 8× v5p-8 — affordable.

Full run — accuracy

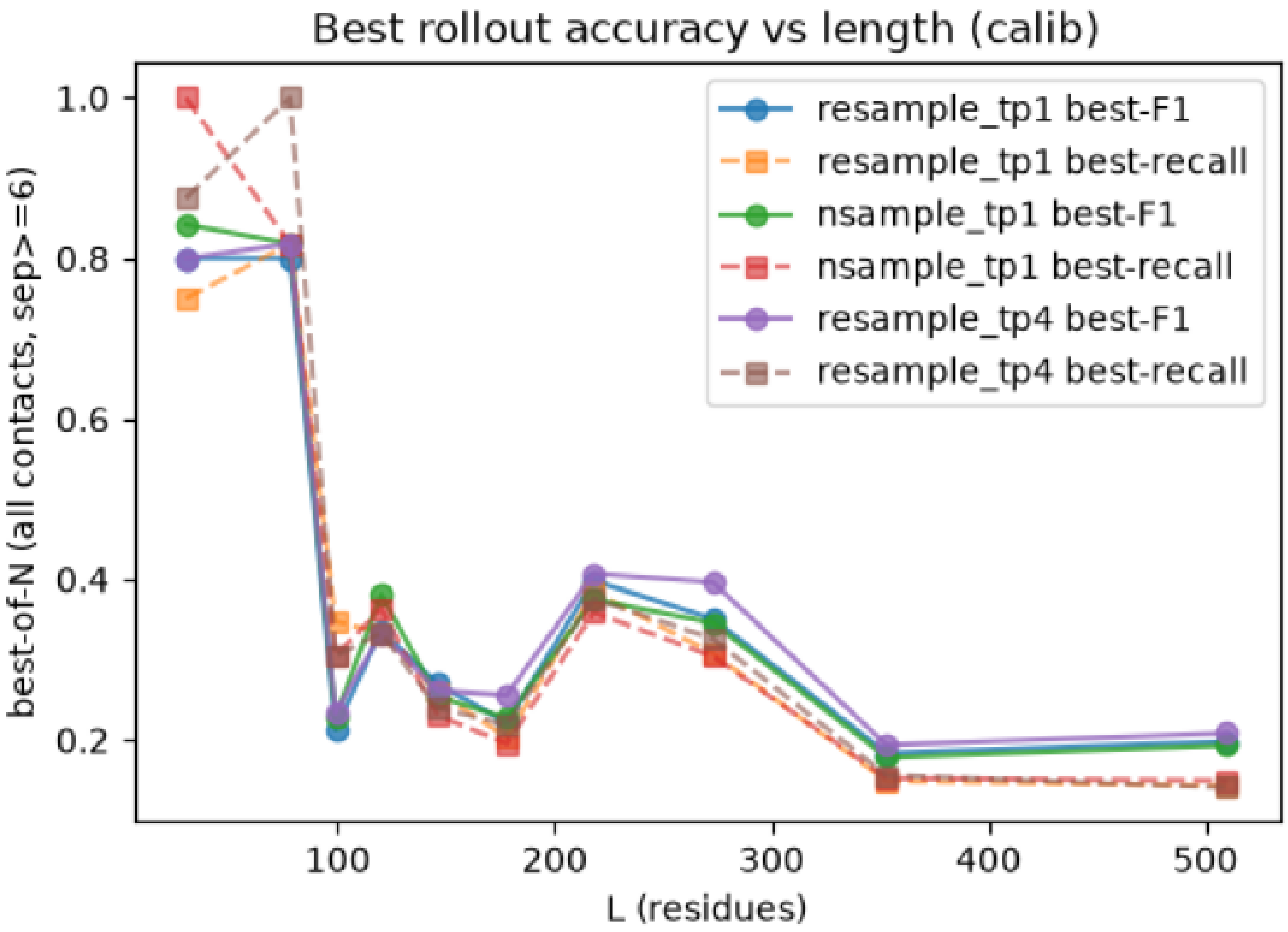
Best-of-1000 per target (all sep≥6): recall mean **0.335** (max 1.0), F1 mean **0.340** (max 0.955) — ~3× the typical single rollout. Strongly length-dependent (best F1 0.52 at L≤100 → 0.20 at L>400); 20% of targets reach F1 ≥ 0.5. Sampling many rollouts surfaces far higher-quality contact sets than any single decode — the raw material for the train-on-rollouts idea.

Deliverables

Per target: precision/recall/F1 of all 1000 rollouts + the best-recall and best-F1 rollouts saved verbatim (complete contacts-v1 documents). Interactive explorer: explore_results.ipynb (Colab).

calib_bestacc_vs_L.png

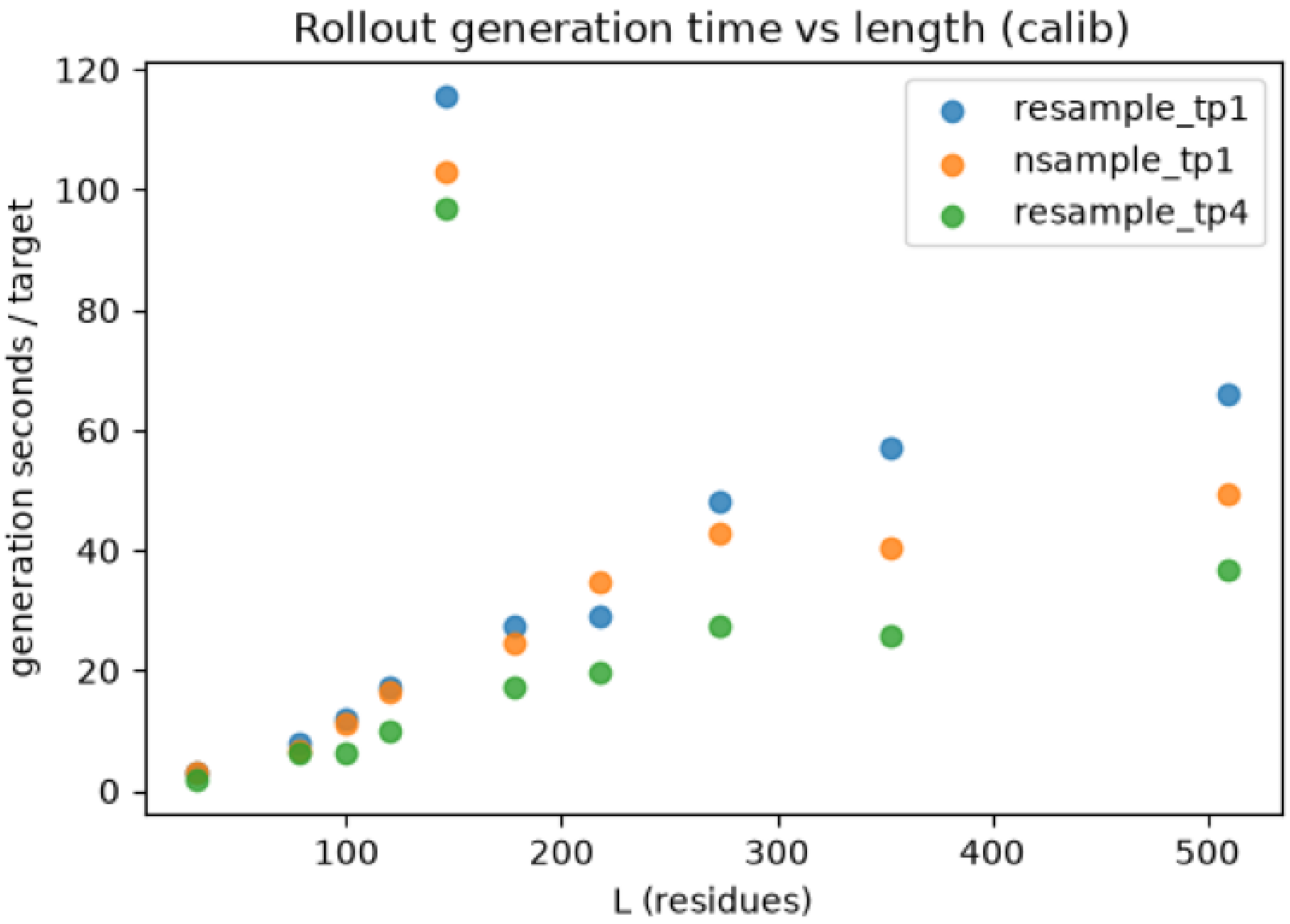
Best-of-N recall / F1 per target vs length.




```
$ aggregate_results.py --runs resample_tp1=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_resample nsample_tp1=gs://marin-us-east5/protein-struct
```

calib_gentime_vs_L.png

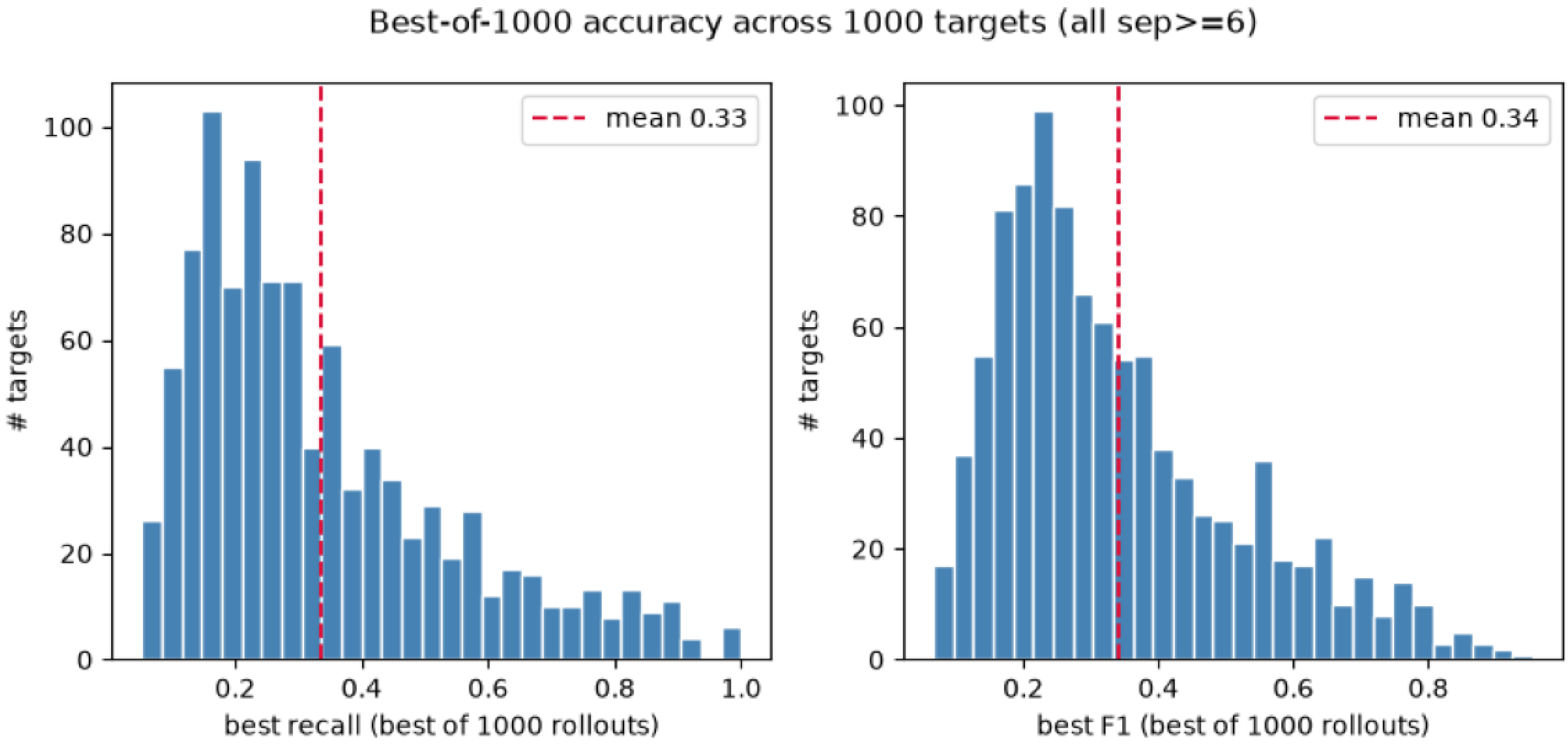
Per-target rollout generation wall time vs protein length.




```
$ aggregate_results.py --runs resample_tp1=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_resample nsample_tp1=gs://marin-us-east5/protein-struct
```

full_best_hist.png

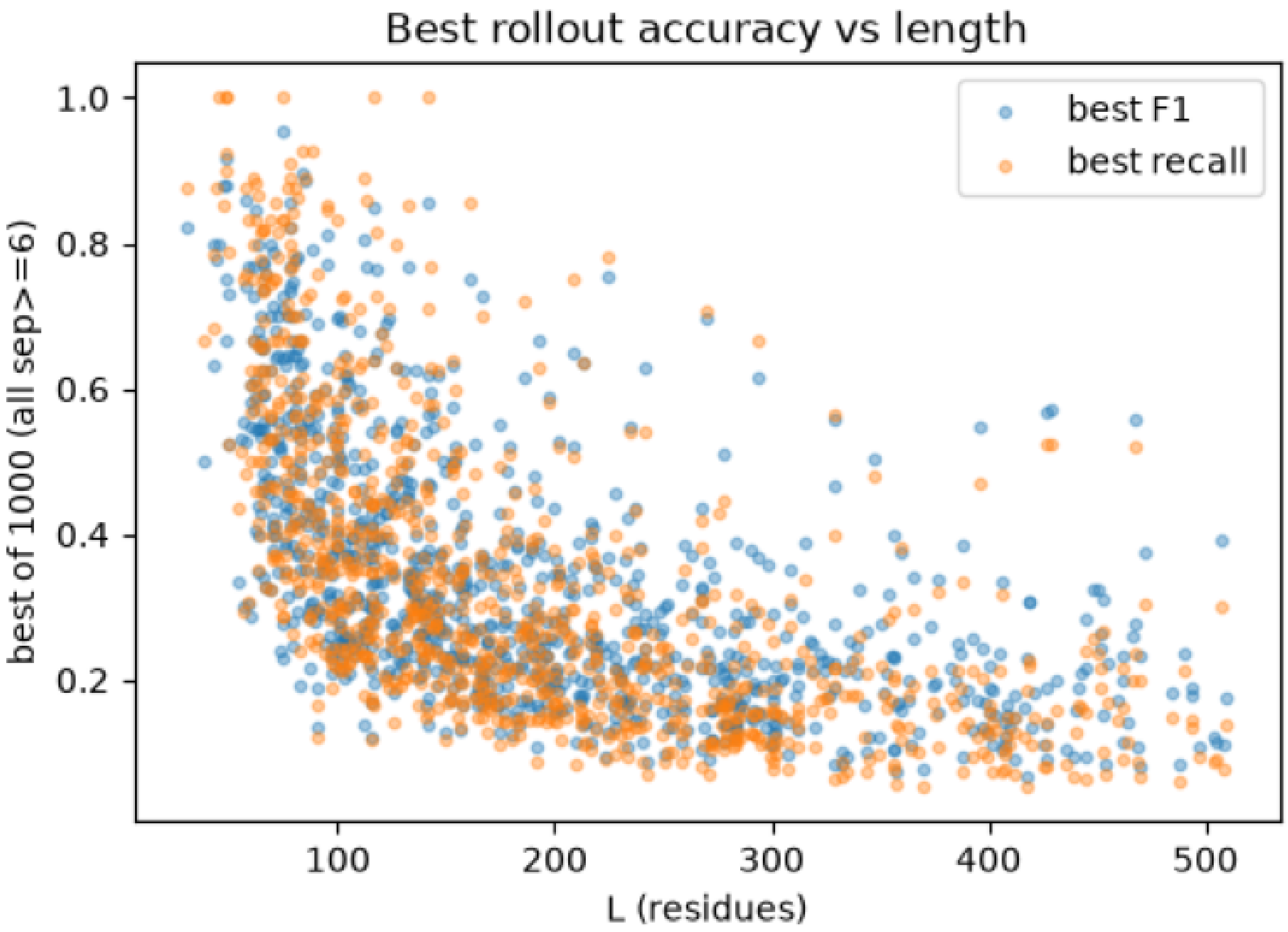
Distribution of per-target best-of-1000 recall and F1 (all contacts, sep>=6).




```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full --gcs-summary gs://marin-us-east5/protein-stru
```

full_bestacc_vs_L.png

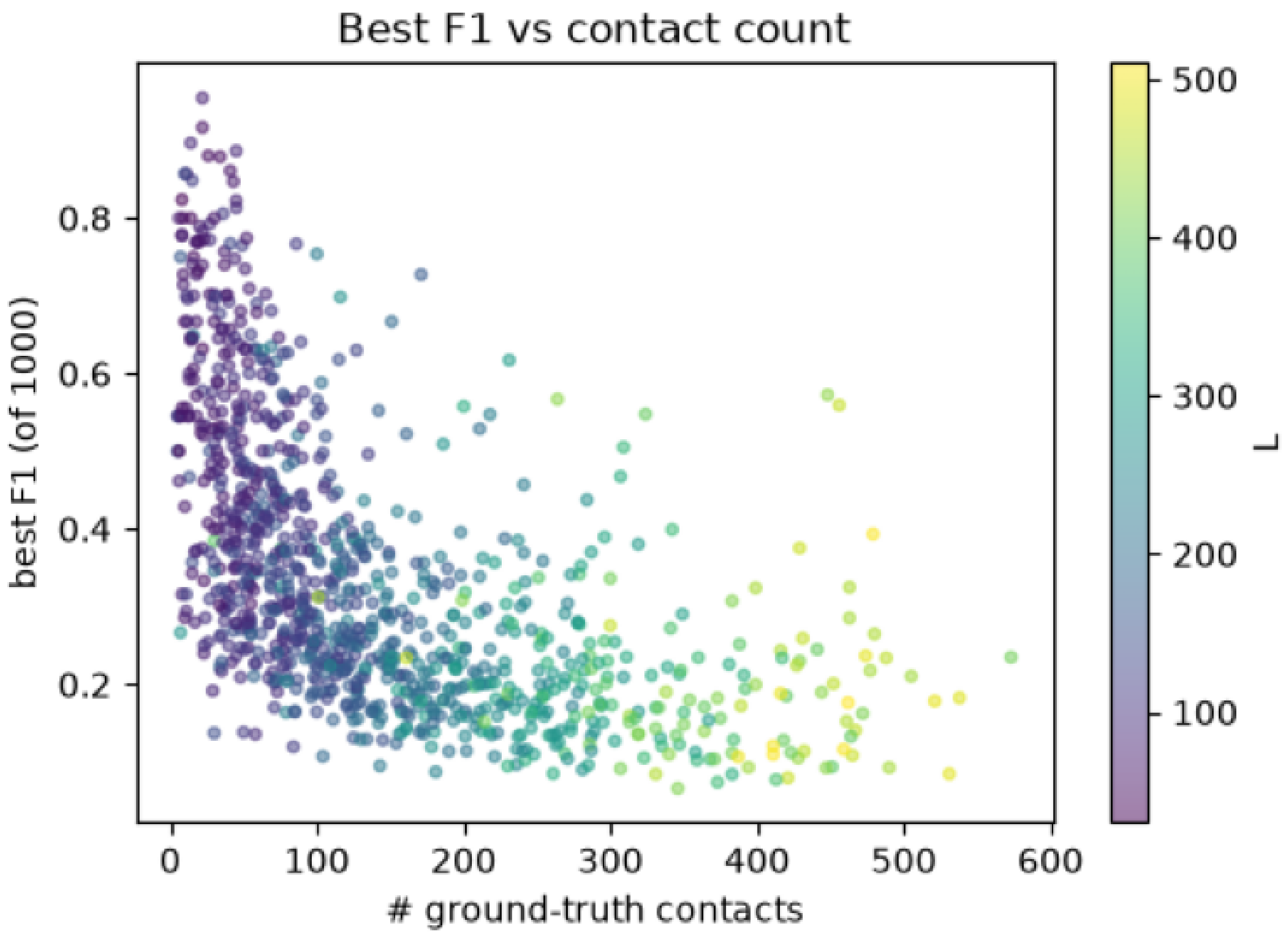
Per-target best-of-1000 recall/F1 vs protein length.




```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full --gcs-summary gs://marin-us-east5/protein-stru
```

full_bestf1_vs_ngt.png

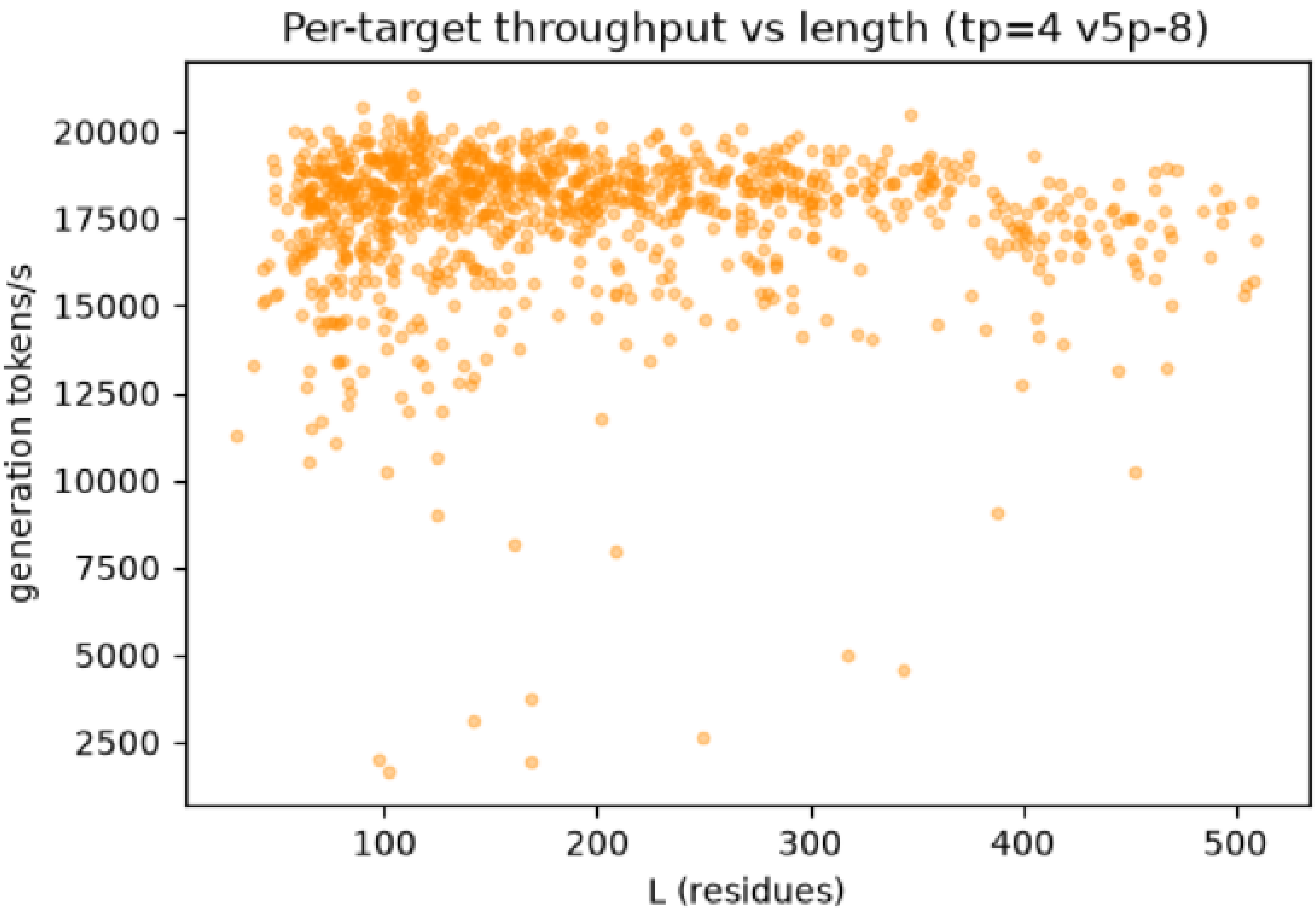
Per-target best-of-1000 F1 vs number of ground-truth contacts (colored by L).




```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full --gcs-summary gs://marin-us-east5/protein-stru
```

full_throughput_vs_L.png

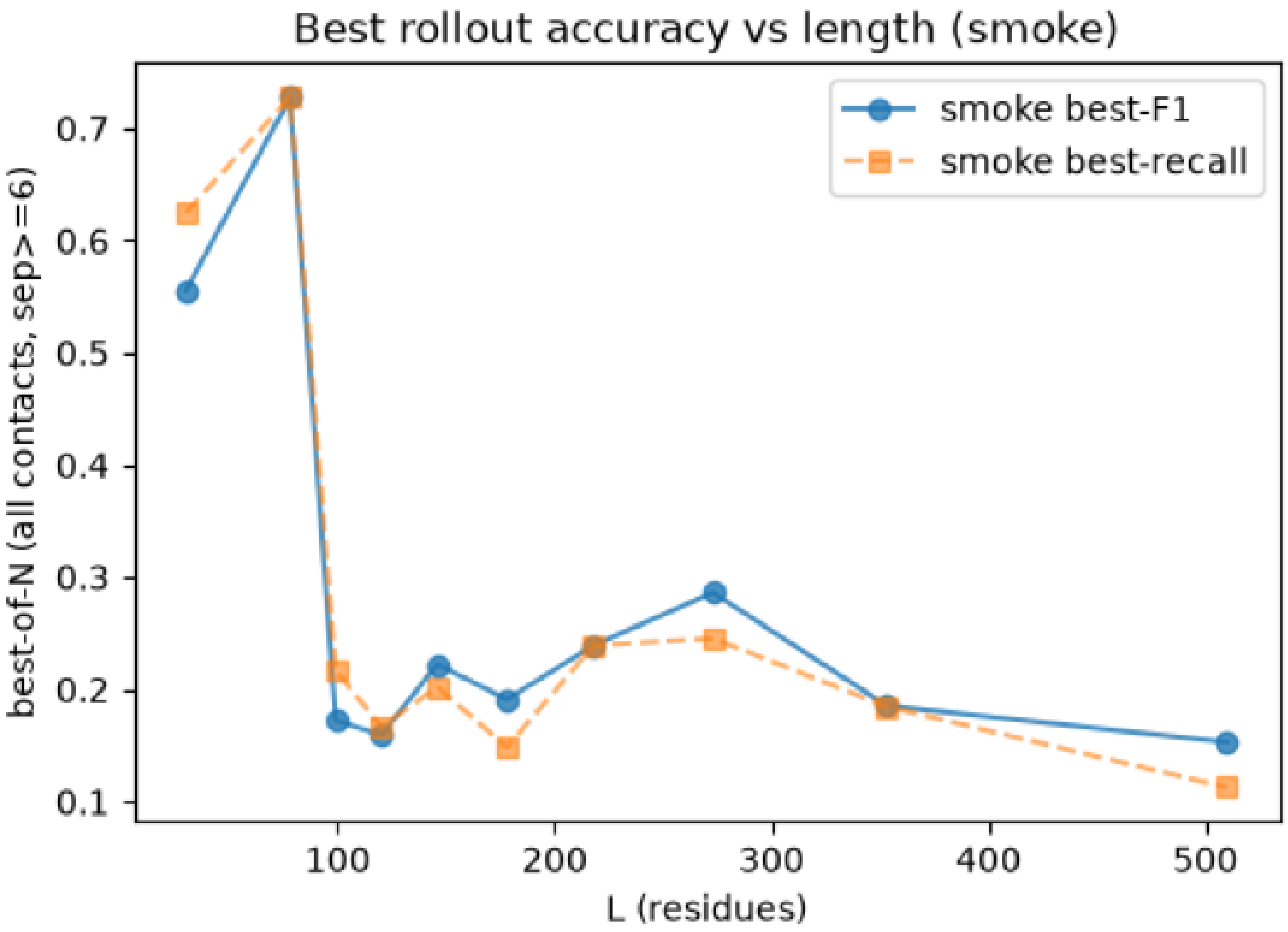
Per-target generation throughput (tokens/s) vs length.




```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full --gcs-summary gs://marin-us-east5/protein-stru
```

smoke_bestacc_vs_L.png

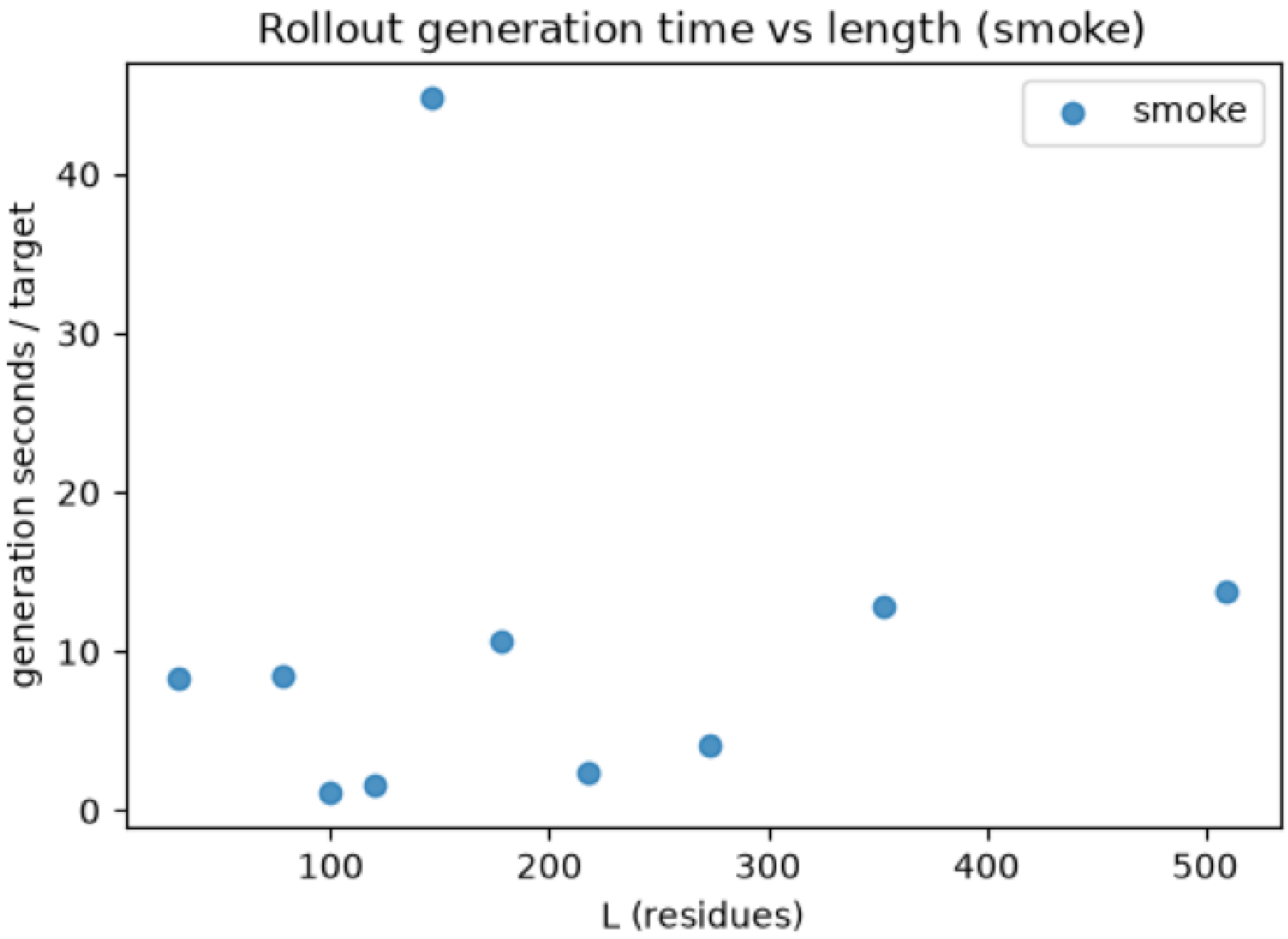
Best-of-N recall / F1 per target vs length.




```
$ aggregate_results.py --runs smoke=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/smoke --out-prefix smoke
```

smoke_gentime_vs_L.png

Per-target rollout generation wall time vs protein length.




```
$ aggregate_results.py --runs smoke=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/smoke --out-prefix smoke
```